

Data sources, completeness, caveats & glossary

A plain reference for the Bournemouth → Brockenhurst noise analysis, so anyone (including the airport or council) can see exactly where the numbers come from, how complete they are, and precisely how each term is defined. Figures are for the three full calendar years **2023-2025** unless stated.

1. Where the data comes from

Every height and position in this analysis ultimately comes from the **aircraft themselves**. Modern airliners continuously broadcast their GPS position, altitude, speed and identity over radio (a system called **ADS-B**). Anyone with a receiver can pick this up — which is what makes this evidence **independent of the airport**.

The chain of custody is:

1. **Aircraft ADS-B broadcast** — the plane transmits its own position and altitude (~ once per second).
2. **OpenSky Network** — a non-profit research network of ground receivers that records these broadcasts. (opensky-network.org)
3. **OPDI v0.0.2** — the Open Performance Data Initiative, run by **EUROCONTROL's Performance Review Unit**, republishes the OpenSky data as clean, checked, downloadable files. This analysis uses OPDI. (opdi.aero)
4. **This analysis** — filters OPDI to arrivals at Bournemouth (EGHH), measures each aircraft's height as it passes Brockenhurst, and compares it to the quiet-descent profile.

Two OPDI tables are used:

- **Flight list** (one row per flight): callsign, aircraft type, origin, destination, date, first/last seen. Used to pick out Bournemouth arrivals and classify the aircraft.
- **Flight events** (points along each flight): each has a time, latitude, longitude and **altitude in feet**, tagged with what happened there — e.g. level-off start/end or crossing 5,000 / 7,000 / 10,000 ft.

Intended cross-check (WebTrak): the airport publishes its own radar/ADS-B tracks on its WebTrak website. The plan is to spot-check a sample of flights (matched by callsign and time) against WebTrak so the airport cannot dismiss the public-data finding without contradicting its own published data. WebTrak has no bulk download, so it is a validation layer, not the mass-data source.

2. What period and how much data

- **Period analysed:** 1 January 2023 → 1 January 2026 (the three full years)

2023, 2024, 2025 are reported).

- **OPDI data availability:** January 2022 to ~January 2026, refreshed periodically. (This is why the June-2026 resident audit dates cannot yet be matched — they are beyond current coverage.)

- **Files processed:** the events come in 10-day files. Of **111** files across the period, **110** were processed; **1 was missing on the server** (the window 2–12 Oct 2025) and one further window returned no Bournemouth records — together about **20 days (~1.8%)** of the three years are absent. This does not materially affect the yearly totals or trends.

3. How complete it is

This is the most important section to read before quoting any single number.

Flights tracked (very high):

Measure (2023–2025)	Value
Total Bournemouth arrivals (all aircraft)	33,157
— of which large passenger jets	11,453
Arrivals with usable trajectory data	32,197 (97%)
Large jets with usable trajectory data	11,091 (97%)

So **almost every arrival is captured** — this is not a small or cherry-picked sample of flights.

Exact height directly over the village (a smaller subset):

Measure (2023–2025)	Value
Arrivals with a precise over-Brockenhurst height	4,938 (15%)
Large jets with a precise over-Brockenhurst height	2,051 (18%)
Night large jets with a precise height	217
Night large jets with a height at/near the village	321

Why the drop from 97% to ~15–18%? The public data records **key moments** along each flight (levelling off, crossing set altitudes), **not** a continuous line.

A precise "over the village" height exists only when one of those recorded moments happens to fall right at Brockenhurst — roughly **1 flight in 6**.

What this means for the numbers:

- The **descent-profile graph** (height vs distance) uses all recorded points in the approach corridor — **99,655 data points** — so it is a large, robust sample.

- The **exact-height counts** ("N airliners below 2,000 ft over the village") are drawn from that ~1-in-6 subset, so they are **minimums**. The true numbers are **higher**, not lower — there is no mechanism that would make them overstated.

4. Caveats & things to be aware of

- **Counts are floors, not ceilings.** Because precise over-village heights exist for ~1 in 6 flights, every "how many were too low" figure understates the real total.
 - **Milestone data, not a continuous track.** OPDI records events (level-offs, altitude crossings), not a second-by-second path. The level-off evidence is robust; the exact-height evidence is a representative subset.
 - **A single low or level flight can be innocent.** Air traffic control may level an aircraft for separation or weather. The argument here is **statistical**: a large and growing share of arrivals lower than a continuous descent is inconsistent with "continuous descent ... at all times", whatever any individual justification.
 - **Altitude is barometric** (referenced to standard pressure below the transition altitude). Any error is tens of feet — small next to the ~1,100 ft gap between where aircraft are and where a quiet descent would put them.
 - **Standard 3° glidepath assumed** for the quiet-descent line. If Bournemouth's instrument glidepath differs, the line shifts slightly; the value is a single configurable setting.
 - **"Large jet" excludes business jets** (they are counted separately). It includes some smaller regional jets; see the glossary.
 - **Landing time** is taken from the last recorded position; **day/evening/night** is then set in **UK local time** (so it follows British Summer Time correctly).
 - **~20 days are missing** (Section 2); yearly trends are unaffected.
 - **Reception thins at very low level** far from receivers, which is part of why close-in points are sparser — but 97% of flights are still captured overall.
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5. Glossary — definitions and values

ADS-B — the radio signal an aircraft broadcasts with its own GPS position, altitude, speed and identity. The raw source of all heights here.

Bournemouth Airport (EGHH) — the arrival airport. Reference point **50.7805°N, 1.8396°W**, elevation **38 ft**. Runway 26 (landing towards the west) is the relevant runway; its landing threshold is the eastern end, nearest Brockenhurst.

Brockenhurst — the New Forest village being overflown, at **50.8189°N, 1.5757°W**. It lies **~10.3 nautical miles** from the airport on a bearing of **~077°**, which is **directly under the runway-26 approach line** (final approach track 255°). It is **~9.8 nautical miles** from the runway threshold. This is why nearly every westerly arrival passes over it.

Quiet descent line / "continuous descent approach" (CDA) — the ideal quiet way to arrive: the aircraft descends steadily with engines near idle, never levelling off. Modelled as a standard **3° glidepath**, which loses about **318 ft per nautical mile**. Extended back from the runway threshold (elevation 38 ft), this puts a compliant aircraft at about **3,116 ft over**

Brockenhurst. **On the graphs this is the orange line; falling below it** means the aircraft is lower than a quiet descent would have it — usually flying low and level under power, which is what makes the noise.

2,000 ft procedure floor — Bournemouth's own published Noise Abatement rule states that turbine-powered / heavier (>5,700 kg) aircraft "**shall not descend below 2000 FT QNH before intercepting the glidepath, nor thereafter fly below it.**" (Source: UK AIP, EGGH AD 2.21, current AIRAC cycle — the official rule-book; also on the Jeppesen chart in `docs/`.) On the graphs this is the **red dashed line**. **Note the airport tolerates down to 2,000 ft before the aircraft joins** the final approach path, so being at ~2,300 ft is not itself a breach — the breach is **level flight** (see below), which the same rules prohibit.

Level-off (non-continuous descent) — a recorded segment where the aircraft stops descending and flies level. A continuous descent has no level flight; to hold level altitude on approach the engines must be pushed up, so a level-off is both a departure from "continuous descent" and a direct cause of extra noise.

"Level-off over the village" = a level segment starting within **6 km** of Brockenhurst.

Over-Brockenhurst height ("gate" height) — the aircraft's measured height as it passes the village, taken from a recorded point within **3 km** of Brockenhurst and within the approach corridor.

Approach corridor — the funnel along the extended runway-26 centreline used to separate genuine arrivals from unrelated traffic: within **4 km** either side of the centreline, on the approach side of the airport.

Aircraft categories — each flight is classified from its **ICAO type code** (e.g. B738 = Boeing 737-800), with the aircraft's engine/type class as a fallback:

- **Large jet** (the "airliners" the case is about) — commercial narrow-body and wide-body jets: Boeing 737 family (incl. MAX), Airbus A320 family, A220, A330/A340/A350/A380, Boeing 757/767/777/787/747, Embraer E-Jets, regional jets (CRJ), and similar. **Excludes business jets.**

- **Business jet** — executive jets (e.g. Falcon, Gulfstream, Citation, Learjet, Challenger, Phenom). Counted separately, not as airliners.

- **Turboprop** — propeller airliners/commuters (e.g. ATR 42/72, Dash 8).

- **GA / light** — small private piston aircraft (e.g. Cessna 172, Piper).

- **Helicopter** — all rotary-wing.

- **Other** — anything not identifiable into the above (e.g. missing type code).

Time-of-day windows — set in **UK local time** at the moment of landing:

- **Day** = 06:00–19:00

- **Evening** = 19:00–23:00

- **Night** = 23:00–06:00 (the core sleep window; the same 7-hour period Heathrow treats as its night quota period)

Large jet (engine-class fallback) — where the exact type code is missing, classification falls back to the ICAO class: multi-engine jet → treated as a

jet, turboprop → turboprop, piston → GA/light, rotary → helicopter.

Everything above is reproducible from the code in this repository. The full technical derivation (geometry, formulae, thresholds) is in `METHODOLOGY.md`; all tunable values live in `config.py`.